

Topic: Client-side Configuration

⇒ Naming methods define the way the simple names

Host naming method: map to connect descriptors

⇒ Clients can connect to a server using a host name if:

- Connecting to an Oracle database service using Oracle net services, client software.

- Client and server are connecting over a TCP/IP protocol.

- Host names are resolved through an IP address translation mechanism such as DNS or local /etc / hosts file.

- No advanced features like connection managers or security options are used.

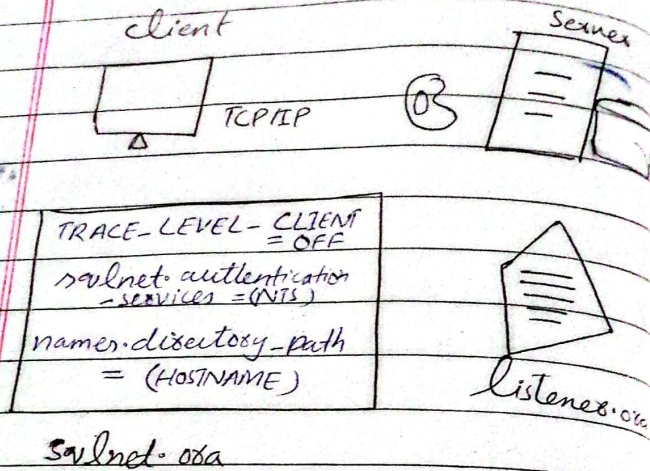
⇒ " Host naming method allows ^{clients} to connect to the db by specifying host name of server where the db resides."

- Limited by the granularity of control and the need for consistent host name across the network.

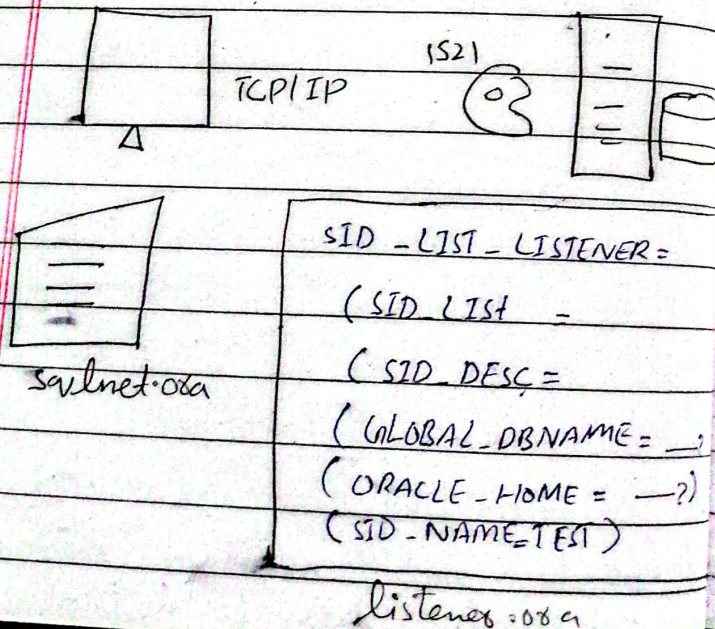
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Host naming client side:



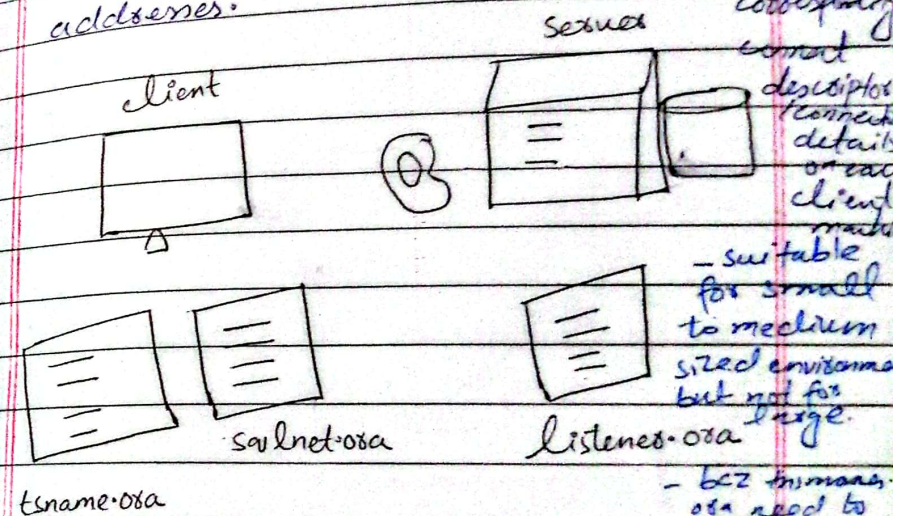
Host naming server side:



Local naming method:

⇒ The local naming method uses a local configuration file, typically 'tnsnames.ora' to map service names to network addresses.

⇒ This method stores DB service names and their corresponding connect descriptors on each client machine.



TNS stands for Transparent network substrate. Tnsnames.ora is a file on client side is use to have a description of connection with a db on server side.

- bcz tnsnames.ora need to update on each machine in case of change.

Topic:Net Configuration AssistantDefinition:

- ⇒ Oracle net configuration assistant
- helps set up the network configurations after installing oracle software.
- ⇒ it creates the initial, basic configuration files automatically.
- ⇒ Oracle net manager tool to manage network services.
- ⇒ These tools have graphical interfaces for configuring the following:
 1. Listeners
 2. Naming methods
 3. Local net services names
 4. Directory usage

Configuring the listener:

⇒ Configuring the listener with ONCA is straightforward. Steps that are followed:

- Select a listener configuration
- Add a listener
- Select a protocol
- Design a port no
- Additional prompt

Naming methods configuration

◦ The Oracle ONCA also help to configure how database services are named.

◦ Some options are:

1. Local (tnsnames.ora file)

⇒ Use a local file to store service name and configuration.

2. Host Names

⇒ Uses TCP naming service.

This method does not support connection pooling as OCI.

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- o Local net service name configured
- o This feature allow to manage net services names:

1. Add:

- o Specify the db version and service name.
- o Enter the global service name or SID.

- o Choose the protocol.
- o Enter the host name and listener port.
- o optionally, test the connection to ensure it works.

Name the new net service.

- o Confirm the creation of local service name.

Delete:

Rename:

Reconfigure:

Test: verify that Configuration information is correct and that a successful connection can be made to the specified db

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- o Directory usage configuration:

Def:

⇒ A directory service stores network information centrally.

Feature of DUC:

- o Store net services names and their location.
- o Provide global db links and aliases.
- o Act as a central repository for client configuration information.
- o Update client configuration file automatically.
- o Store client information like usernames and passwords.

Configuration steps:

⇒ Select the directory server, enter host name, port (default 389), and ssl port default (636).

⇒ Verify or create the required schema and content with appropriate privileges.

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Usage and Configuration of Oracle Shared Server:

- ⇒ Whenever user want to connect to database, it send request to the listener and listener spawns or start new server process and user keep communicating with server process.
- ⇒ Instead of creating new dedicated server process, already create or configure two components:
 - Dispatchers
 - Shared server

Key concepts:

Dispatcher process: (man 10 server or connection)

- ⇒ responsible for managing connections
- ⇒ Not process query
- ⇒ get requests from clients and puts it into request queue

⇒ Oracle shared server a configuration option that allows multiple user processes to share a relatively small number of server process that reduce resource usage and

- ⇒ Any server process who is free picked request from request queue.

Virtual circuits:

- ⇒ Communication pathway between dispatcher and shared server.

Queues tables:

- ⇒ maintain requests and responses between dispatcher and shared servers.

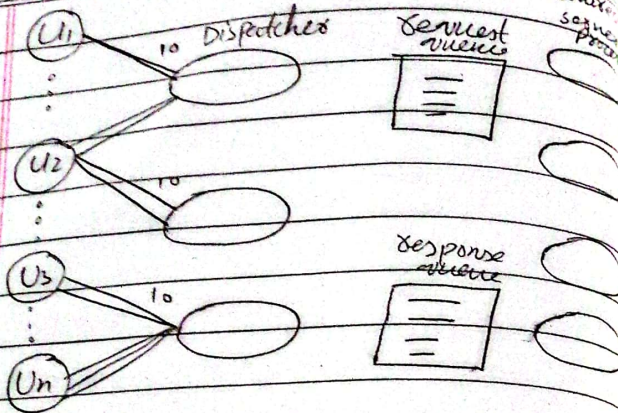
Listener:

- ⇒ Redirects incoming connection requests to dispatcher.

Note:

- if number of shared server processes $<$ no of requests then delay time and que pilling occur.
- if number of shared server processes $>$ no of requests then resource wastage.

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Configure steps:

Enable shared servers:

- o ALTER SYSTEM SET SHARED_SERVERS=1;

Configure dispatcher:

- o ALTER SYSTEM SET DISPATCHER =
(PROTOCOL = TCP)
(DISPATCHER = 3);

Optional parameters:

- o ALTER SYSTEM MAX_SHARED_SERVERS=1;
- o ALTER SYSTEM SHARED_SERVER_SESSIONS=100;
- o ALTER SYSTEM CIRCUITS=200;

Monitor shared servers:

- o V\$SHARED_SERVER: Display server info.
- o V\$DISPATCHER: Display dispatcher info.
- o V\$CIRCUIT: Display info of virtual circuits.

Usage / Benefits of Shared Servers:

⇒ Idea of shared server processes enable to optimize server side processes in terms of resource utilization such as PGA, CPU etc. if dedicated server process use then each process has PGA that cause memory overhead.

1. Reduced memory usage:

- o fewer server process mean less memory overhead per user connection.

2. Scalability: Can handle many requests compared to dedicated process.

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3. Improved performance:

- o Better utilization of server resources.